SO DO KHOI v3 – ESP32-H2 + pin CR123A + module do am dat dien dung (Matter over Thread)

CR123A 3.0V
Li-MnO2 ~1500mAh
(de pin co day)
J3: 2 pad han + / -

VBAT_RAW

Chong nguoc cuc
Q1 AO3415A
(P-MOSFET)

VBAT

Buck-boost TPS63900
1.8-5.5V -> 3.3V
Iq 75nA, >90% @10uA

+3V3

ESP32-H2-WROOM-02C
RISC-V 96MHz, 802.15.4 + BLE
Matter over Thread
SED / ICD (SIT/LIT)

IO1 ADC1_CH0 SOIL_ADC
IO2 ADC1_CH1 VBAT_ADC
IO10 SENS_PWR_EN_N
IO11 VBAT_MEAS_EN
IO9 nut BOOT/USER
IO0 LED
IO23/24 UART0, IO26/27 USB

UI / nap:
SW1 BOOT/USER (IO9)
LED do (IO0)
J1: 8 pad tron (UART+USB)

Anten PCB cua module
nho ra ngoai mep PCB
khong dong ben duoi

Do pin: Q2/Q3 cong tac
+ cau chia 1/2
(0 ro khi ngu)

VBAT_ADC

Load switch Q4 (P-FET)
-> VSENS (chi bat khi do)

SENS_PWR_EN_N

VSENS

J4 PH2.0-3P ->
MODULE DO AM DAT DIEN DUNG
Capacitive Soil Moisture v1.2
(TLC555), AOUT 0-3V, ~5mA

D3 TVS: chong ESD day AOUT

Power

Power: CR123A,
RPP,
TPS63900 3V3,
VBAT sense,
VSENS

File: power.kicad_sch

MCU

SOIL_ADC

MCU: ESP32-H2-WROOM-02C (Thread ICD)

File: mcu.kicad_sch

Sensors

Soil module connector: J4 PH2.0-3P,
ESD,
ADC filter

File: sensors.kicad_sch

UI

UI: button,
LED,
programming pads

File: ui.kicad_sch

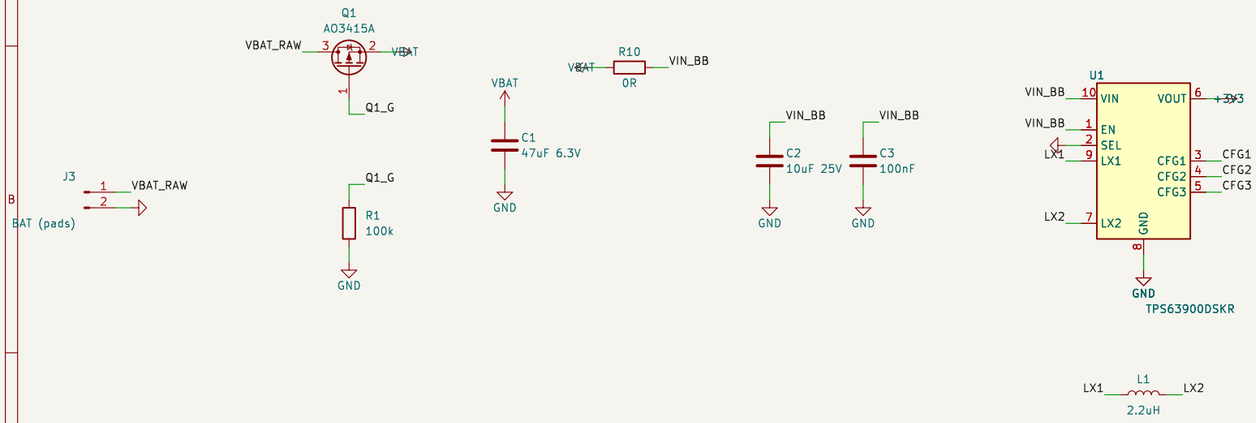
Root sheet: so do khoi + 4 sheet con (ket noi bang global label)
H2 Soil Sensor

Sheet: /
File: h2-soil-sensor.kicad_sch

Title: H2 Soil Sensor – Matter over Thread (ESP32-H2, CR123A)

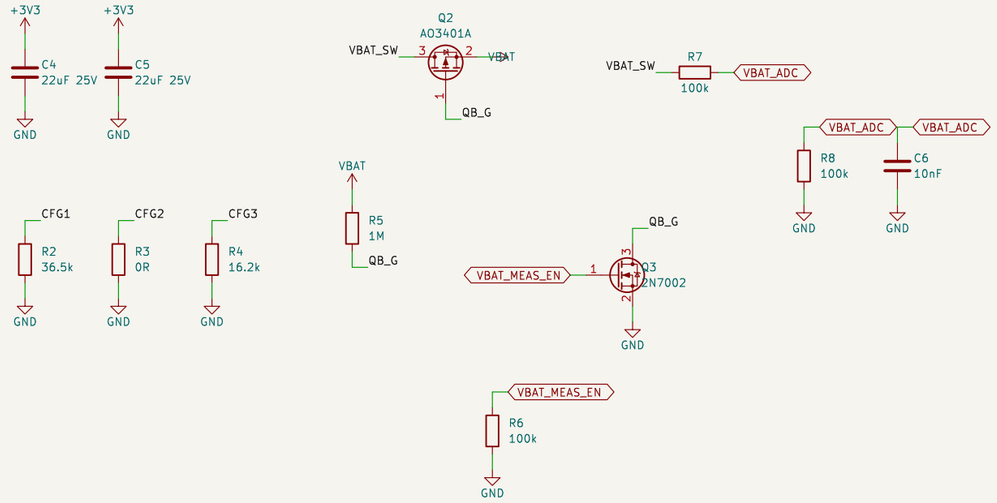
Size: A3	Date: 2026-11-12	**Rev: 3.0**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 1/5

1) PIN + CHONG NGUOC CUC
Day cua de pin CR123A han vao 2 pad J3 (+ vuong, – tron).
Q1 P–MOSFET: D–>pin(+), S–>tai, G–>GND qua R1.
Lap nguoc pin: Vgs=0 –> khong co dong. R10 OR: cau do dong.

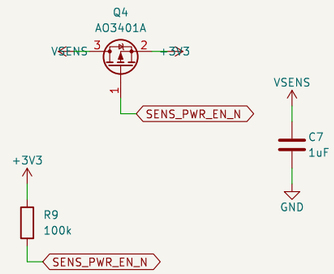


2) TPS63900 BUCK–BOOST –> 3.3V (Iq 75nA)
SEL=GND –> VO(1) theo CFG3=16.2k = 3.3V
CFG1=36.5k (VO(2)=3.3V), CFG2=0R: lin khong gioi han
R CFG 1%/100ppm, khong dat tu tren chan CFG.
L1 2.2uH Isat>=2A; COUT 2x22uF; CIN 10uF.

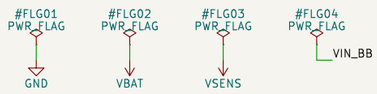
3) DO DIEN AP PIN (khong ro khi ngu)
VBAT_MEAS_EN=1 –> Q3 ON –> Q2 ON –> VBAT/2 –> IO2.
Khi ngu: R6 giu Q3 OFF, R5 giu Q2 OFF.



4) CONG TAC NGUON MODULE DO AM DAT (VSENS)
SENS_PWR_EN_N=0 –> Q4 ON: cap 3.3V cho module.
Firmware bat mem (pull–down noi ~1ms) de han dong nap tu.
R9 giu Q4 OFF khi GPIO Hi–Z (sleep/reset).



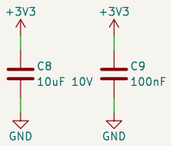
PWR_FLAG (cho ERC)

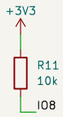


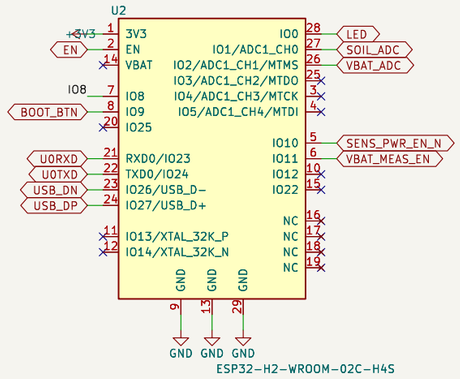
ESP32–H2–WROOM–02C, CR123A, TPS63900
H2 Soil Sensor – Matter over Thread
Sheet: /Power/
File: power.kicad_sch
Title: Power: CR123A, RPP, TPS63900 3V3, VBAT sense, VSENS

Size: A3	Date: 2026–11–12	**Rev: 3.0**
KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4		Id: 2/5

ESP32-H2-WROOM-02C (4MB: -H4S de co OTA). Vdd 3.0-3.6V.
Module da co RC tren EN (10k/1uF).
Strapping: IO9=1 (R21, sheet UI) -> SPI boot; IO8=1 (R11).
IO10/IO11 la LP GPIO: giu muc khi light-sleep tat mien TOP.
ADC1: IO1=SOIL (module), IO2=VBAT. Chan khac de trong.





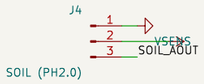


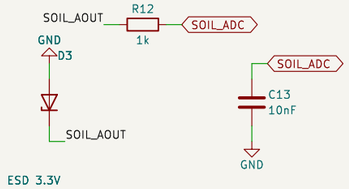
Anten PCB: module o mep tren, vung anten (6mm) nho ra ngoai PCB;
khong dong/GND/linh kien duoi va quanh anten; vo hop cach anten >=15mm.

ESP32-H2-WROOM-02C, CR123A, TPS63900
H2 Soil Sensor – Matter over Thread

Sheet: /MCU/		
File: mcu.kicad_sch		
Title: MCU: ESP32-H2-WROOM-02C (Thread ICD)		
Size: A3	Date: 2026-11-12	**Rev: 3.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/5

MODULE DO AM DAT DIEN DUNG 'Capacitive Soil Moisture Sensor v1.2'
J4 PH2.0-3P (cung loai cong tren module): 1=GND 2=VCC 3=AOUT.
Nuoi tu VSENS 3.3V, chi bat khi do. Chon module dung TLC555 (NE555 khong chay o 3.3V).
AOUT 0-3V (kho = cao) -> R12 1k -> IO1 (ADC1, ATTEN 12dB); C13 loc; D3 TVS chong ESD.







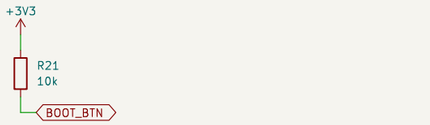
ESP32-H2-WROOM-02C, CR123A, TPS63900
H2 Soil Sensor – Matter over Thread

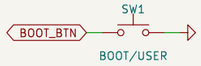
Sheet: /Sensors/
File: sensors.kicad_sch

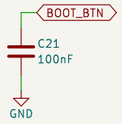
Title: Soil module connector: J4 PH2.0-3P, ESD, ADC filter

Size: A3	Date: 2026-11-12	**Rev: 3.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/5

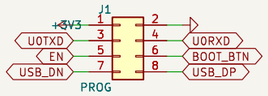
SW1 (IO9/BOOT): 1 lan = do & bao cao ngay / mo commissioning:
2 lan = hieu chuan KHO; 3 lan = hieu chuan UOT; giu >=10s = factory
reset. Giu SW1 khi cap nguon/reset = che do nap (download).











J1: 8 pad tron SMD 2x4 buoc 2.54mm (pogo / han day)
1=3V3 2=GND 3=TXD0 4=RXD0 5=EN 6=IO9 7=USB_D- 8=USB_D+
USB-Serial/JTAG: lap pin + noi D+/D-/GND (khong noi 3V3).
UART: thao pin, cap 3V3 tu adapter USB-UART 3.3V.

ESP32-H2-WROOM-02C, CR123A, TPS63900
H2 Soil Sensor – Matter over Thread

Sheet: /UI/		
File: ui.kicad_sch		
Title: UI: button, LED, programming pads		
Size: A3	Date: 2026-11-12	**Rev: 3.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/5